Decentralized Knowledge Verification: Claims, Inference, and Factchecking with Epistemic Mesh

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Abstract

We present a decentralized knowledge verification system implemented as 8 Holochain zomes: claims registration, knowledge graph construction, semantic query, logical inference, multi-source factchecking, prediction market integration, distributed key generation for knowledge attestation, and cross-cluster bridge. The system forms an "epistemic mesh" where knowledge claims are verified through multiple independent pathways—peer review, logical inference, prediction market pricing, and cross-domain corroboration—each contributing to a composite confidence score. Verification quality is weighted by the verifier's consciousness tier and domain engagement, creating an incentive structure where epistemic authority derives from demonstrated knowledge rather than institutional position. The knowledge graph is distributed across Holochain's DHT with typed relationships (supports, contradicts, extends, specializes) and supports transitive confidence propagation across claim chains. Integration with 15 other Mycelix clusters enables cross-domain knowledge verification: a climate claim can be corroborated by supply chain provenance data, space-based observations, and energy grid measurements.

1 Introduction

Centralized knowledge systems—Wikipedia, Snopes, institutional peer review—face a fundamental trust problem: who verifies the verifiers? Centralized platforms concentrate epistemic authority in editors, administrators, and reviewers whose incentives may not align with truth.

Decentralized alternatives have emerged—prediction markets for forecasting, blockchain-based fact-checking protocols, and peer-to-peer review systems—but these typically address only one aspect of knowledge verification. A claim about climate change requires scientific evidence, statistical analysis, observational data, and economic context. No single verification mechanism suffices.

We present the Mycelix Knowledge cluster: an *epistemic mesh* where multiple verification pathways converge on composite confidence scores for knowledge claims.

2 Architecture

2.1 Seven Zomes plus Bridge

1. **claims**: Structured claim registration with provenance metadata, evidence links, domain tags, and confidence scoring.
2. **graph**: Knowledge graph construction with typed relationships between claims.
3. **query**: Semantic query processing across the distributed knowledge graph.
4. **inference**: Logical inference engine for deriving new claims from existing premises.
5. **factcheck**: Multi-source fact verification with consciousness-weighted confidence aggregation.
6. **markets_integration**: Prediction market integration for epistemic value discovery.
7. **bridge**: Cross-cluster knowledge dispatch.

2.2 Data Model

Claims are the atomic unit of knowledge:

```
pub struct Claim {
    pub statement: String,
    pub domain: Vec<String>,
    pub evidence: Vec<EvidenceLink>,
    pub confidence: f64,  // [0, 1]
    pub author: AgentPubKey,
    pub created_at: Timestamp,
    pub schema_version: u8,
}

pub struct EvidenceLink {
    pub source: EvidenceSource,
    pub uri: String,
    pub strength: f64,  // [0, 1]
}

pub enum EvidenceSource {
    DirectObservation,
    PeerReview,
    StatisticalAnalysis,
    ExpertTestimony,
    InstrumentMeasurement,
    CrossDomainCorroboration,
}
```

3 Knowledge Graph

3.1 Typed Relationships

The graph zome maintains a typed knowledge graph where nodes are claims and edges are relationships:

Definition 1 (Knowledge Edge Types). *1.* ***Supports****: Claim A provides evidence for Claim B. Strength: how strongly A supports B.*

2. ***Contradicts****: Claim A provides counter-evidence to Claim B. Strength: how strongly A undermines B.*
3. ***Extends****: Claim A generalizes or broadens Claim B.*
4. ***Specializes****: Claim A is a specific instance of Claim B.*
5. ***Requires****: Claim A is a necessary premise for Claim B.*
6. ***Supersedes****: Claim A replaces an outdated Claim B.*

3.2 DHT Distribution

The knowledge graph is distributed across Holochain's DHT:

- Claim entries are hashed and stored in their DHT neighborhood
- Link entries (relationships) are attached to source claim hashes
- Query processing traverses links across DHT neighborhoods
- Validation rules ensure relationship consistency (e.g., a Supersedes edge requires both claims to share at least one domain tag)

3.3 Transitive Confidence Propagation

Confidence propagates through the knowledge graph along Support edges:

$$c_{\text{derived}}(B) = \max\left(c_{\text{direct}}(B),\ \max_{A \to_s B} c(A) \cdot s(A,B) \cdot d^{h(A,B)}\right) \tag{1}$$

where $s(A,B)$ is the support strength, $d \in (0,1)$ is the per-hop decay, and $h(A,B)$ is the hop distance. Contradiction edges reduce confidence:

$$c_{\text{adjusted}}(B) = c_{\text{derived}}(B) \cdot \prod_{A \to_c B} (1 - c(A) \cdot s_c(A,B)) \tag{2}$$

This formulation ensures that a single strong contradiction can significantly reduce confidence, while weak contradictions have proportionally small effects.

4 Claim Lifecycle

4.1 Registration

Any Participant+ tier agent can register a claim. The claim entry is validated by the integrity zome:

1. Statement is non-empty and within length limits
2. At least one domain tag is provided
3. Author's consciousness credential is valid
4. Schema version is current

4.2 Verification

Independent agents verify claims by submitting verification entries:

```
pub struct Verification {
    pub claim_hash: ActionHash,
    pub verdict: Verdict,
    pub evidence: Vec<EvidenceLink>,
    pub confidence: f64,
    pub verifier: AgentPubKey,
}

pub enum Verdict {
    Confirmed,
    Disputed,
    Inconclusive,
    NeedsMoreEvidence,
}
```

Verification quality is weighted by the verifier's consciousness profile:

$$w_j = E_j \cdot \text{tier_weight}(\text{tier}_j) \tag{3}$$

where E_j is the verifier's engagement in the claim's domain and tier_weight maps consciousness tiers to weights. This ensures that domain experts with strong governance standing contribute more to the consensus.

4.3 Inference

The inference zome derives new claims from existing premises using logical rules:

```
pub struct InferenceRule {
    pub premises: Vec<ClaimPattern>,
    pub conclusion: ClaimTemplate,
    pub confidence_function: ConfidenceFn,
}
```

Inference runs periodically (not in real-time), scanning the knowledge graph for premise patterns and generating derived claims with appropriate provenance metadata. Derived claims are marked as inference-generated and link back to their premises via Requires edges.

4.4 Factchecking

The factcheck zome aggregates verification results:

$$c_{\text{fact}}(B) = \frac{\sum_{j:v_j=\text{Confirmed}} w_j \cdot c_j - \sum_{j:v_j=\text{Disputed}} w_j \cdot c_j}{\sum_j w_j} \tag{4}$$

Claims with $c_{\text{fact}} > 0.8$ are classified as "well-supported." Claims with $c_{\text{fact}} < -0.5$ are classified as "disputed." Intermediate values indicate ongoing verification.

5 Prediction Market Integration

The markets_integration zome connects knowledge claims to prediction markets:

1. A prediction market is created for a specific claim (e.g., "Global average temperature will exceed 1.5C above pre-industrial by 2030")
2. Participants stake MYCEL on outcomes
3. Market prices reflect collective confidence in the claim
4. Resolved markets adjust participant reputation (correct predictions increase reputation, incorrect predictions decrease it)

Market prices provide an independent signal for claim confidence, complementing the expert verification pathway. The two signals are combined with configurable weights:

$$c_{\text{composite}} = \alpha \cdot c_{\text{fact}} + (1 - \alpha) \cdot c_{\text{market}} \tag{5}$$

where α is the domain-specific weight for expert vs. market confidence (default 0.7, prioritizing expert verification).

6 Cross-Domain Corroboration

6.1 The Epistemic Mesh

The Knowledge cluster's bridge zome enables cross-domain corroboration:

- A climate claim about deforestation can be corroborated by the Space cluster's satellite observations
- A supply chain provenance claim can be verified against the Energy cluster's grid measurements
- A health claim about nutrition can be checked against the Commons cluster's food production data

Cross-domain corroboration uses the `CrossDomainCorroboration` evidence source type and adds Support/Contradict edges to the knowledge graph.

6.2 Cross-Cluster Query

The query zome supports cross-cluster queries via the bridge:

```
pub struct CrossClusterQuery {
    pub source_domain: String,
    pub target_cluster: CrossClusterRole,
    pub query: SemanticQuery,
    pub max_depth: u32,
    pub min_confidence: f64,
}
```

Queries are consciousness-gated: cross-cluster queries require Citizen+ tier to prevent abuse of the bridge dispatch mechanism.

7 Consciousness-Weighted Epistemic Authority

The key innovation of the epistemic mesh is that epistemic authority derives from the consciousness profile rather than institutional position:

Proposition 1 (Epistemic weight decomposition)**.** *An agent's epistemic weight for a claim in domain d is:*

$$w = \underbrace{E_d}_{engagement} \cdot \underbrace{f(tier)}_{governance} \cdot \underbrace{R}_{reputation} \cdot \underbrace{(1 + \beta \cdot h_d)}_{history} \tag{6}$$

where E_d is domain engagement, $f(tier)$ is the tier weight function, R is reputation, h_d is the agent's verification accuracy history in domain d, and β is the history bonus factor.

This decomposition means that an agent's epistemic authority in a domain is proportional to their actual engagement and track record, not their credentials or institutional affiliation.

8 Integration with Symthaea

For agents connected to the Symthaea consciousness engine, the Knowledge cluster provides a bidirectional integration:

- **Knowledge → Consciousness**: Verified knowledge claims feed the Knowledge Manager in Symthaea's cognitive loop, providing grounding for the reasoning engine.
- **Consciousness → Knowledge**: Symthaea's inference outputs (from the reasoning engine) can be registered as claims with appropriate provenance metadata, contributing to the collective knowledge graph.

9 Evaluation

The Knowledge cluster includes:

- Unit tests for claim validation, graph operations, and inference rules
- Integration tests for cross-zome query processing
- Property tests for confidence propagation bounds
- Cross-cluster integration tests for corroboration pathways

10 Related Work

Google Knowledge Graph [1]: Centralized, proprietary, no verification mechanism.

Wikidata [2]: Open but centrally administered. No confidence scoring or prediction market integration.

The Graph [3]: Decentralized indexing for blockchain data. Focuses on data availability, not knowledge verification.

Augur [4]: Decentralized prediction markets on Ethereum. Addresses price discovery but not structured knowledge verification.

Our contribution is the integration of structured claims, logical inference, expert verification, and prediction markets into a single epistemic mesh with consciousness-weighted authority.

11 Conclusion

The epistemic mesh demonstrates that knowledge verification can be decentralized without sacrificing rigor. Multiple independent

verification pathways—peer review, logical inference, market pricing, and cross-domain corroboration—converge on confidence scores that reflect genuine epistemic status rather than institutional authority. Consciousness-weighted verification ensures that domain experts with demonstrated engagement contribute more to the collective knowledge base, while the open participation model (Participant+ for claim registration) ensures that no gatekeeper can suppress inconvenient truths.

References

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